



Analysis of Factors Influencing Regional Economic Development in Shandong Province

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Abstract: As globalization deepens and the digital economy advances, Shandong Province, one of China's major economic regions, is undergoing shifts in its population structure alongside changes in its economic growth drivers. Exploring the impacts of education support, technological progress, and population aging on regional economic development is of great significance for formulating scientific regional economic policies. Based on macro-level data for Shandong Province from 2004 to 2022, this study constructs a double-logarithmic regression model. The ordinary least squares (OLS) method is used to estimate the model parameters, and stepwise regression is applied to address the multicollinearity problem. The empirical results show that fiscal education expenditure is significantly and positively associated with regional gross domestic product (GDP), suggesting that education support may play an important role in promoting regional economic development. Population aging also shows a significant positive association with regional economic development in the corrected model, indicating that the silver economy and the continued utilization of older adults' human capital may provide new development opportunities. However, the direct effect of technological progress, measured by technology market turnover, is not fully verified in the current model. The findings suggest that Shandong Province should continue to prioritize education development, improve elderly-care industries and flexible retirement mechanisms, and transform demographic challenges into potential drivers of regional development.

Keywords: Shandong Province; Regional economic development; Education support; Population aging; Double-logarithmic regression model

1 Introduction

Under the background of globalization and digital economy, Shandong Province, as a major coastal economic province in eastern China, is experiencing continuous transformation and upgrading of its economic growth drivers. With changes in the distribution of educational resources, accelerated technological progress, and the continuous evolution of the population structure, Shandong Province is facing both unprecedented development opportunities and challenges [1]. These macro-level changes have a profound impact on the development model and governance capacity of cities in the province, such as the urban-rural and inter-regional distribution of educational resources, the transformation of technological innovation into industrial advantages at regional and urban levels, and the pressure of population aging on urban public service systems. To analyze how these factors jointly influence regional economic development of Shandong Province, this study uses a multivariate regression model focusing on three core variables: education support, technological progress, and the elderly dependency ratio. The aim is to reveal their mechanisms and pathways of influence on the regional economic development of Shandong Province.

Education support, as a key factor in human capital accumulation, technological progress, as an engine of economic growth, and the elderly dependency ratio, as an important indicator of demographic change, together constitute important dimensions affecting regional economic development of Shandong Province. Through quantitative analysis of the relationship between these variables, this study seeks to provide a scientific basis for economic policymaking in Shandong Province. In particular, the findings may provide decision-making references for urban managers in public resource allocation, industrial planning, and responses to social structural change, thereby promoting the transformation and upgrading of the economic growth model and ensuring long-term economic stability, social harmony, and sustainable development.

2 Literature Review

This study examines the impact of education support, technological progress, and population aging on regional economic development. Its theoretical basis is mainly based on human capital theory, endogenous growth theory, life cycle model and induced technological change theory. This section reviews these core theories and the relevant literature in order to construct the analytical framework of the study.

2.1 Education Support and Regional Development Level

The quality and structure of the population have always been important factors affecting economic growth [2]. Since the reform and opening up, the rapid growth of China's economy has mainly benefited from the rapid growth of capital. However, in today's increasingly competitive global economy, competition among countries has shifted from capital-based competition to human-resource-based competition, and competition in the quality of human resources is essentially competition in education [3].

As early as the 1960s, Schultz and Becker developed human capital theory, which emphasizes education and training as forms of investment in human productive capacity. This theory regards education as a future-oriented investment that accumulates knowledge and skills, improves workers' productivity, and forms human capital [4, 5]. However, a recent study has also critically examined human capital theory, arguing that it may place excessive responsibility on individual workers under conditions of employment insecurity and labor market flexibilization [6].

The accumulation of human capital is a key endogenous driving force for long-term economic growth. Based on the experience of economically developed countries, human capital and scientific and technological support are important pathways affecting economic development [7]. As the economy enters a stage of high-quality development, China's economic growth model should shift toward an education-supported model, emphasizing the central role of education in qualitative competition. Government financial expenditure on education is an important way to improve the overall human capital level of society. Therefore, it is necessary to pay attention to the rational planning of material capital investment and human capital investment in higher education for long-term high-quality economic development [8]. Increasing education expenditure, expanding access to education, improving education quality, building a high-quality talent team, and enhancing the contribution of higher-education human capital can all promote regional economic development. This leads to the following hypothesis:

H1: Educational support has a positive effect on the level of regional development.

2.2 Technological Progress and Regional Development Level

Traditional neoclassical growth theories, such as the Solow model, regard technological progress as an exogenous black box and cannot fully explain its source [9]. The endogenous growth theory, especially the model of Romer and Lucas, treats technological progress as an endogenous factor resulting from economic activities such as research and development (R&D) investment, knowledge spillovers, and human capital accumulation [10]. It drives sustained economic growth by improving total factor productivity and optimizing industrial structure.

Economic growth driven by technological progress can promote structural changes in the economy [11]. It is not only the result of economic structure transformation and upgrading, but also the reason for further economic structure transformation. In the process of industrialization, technological progress is regarded as a core driving force of industrial structure change. It promotes industrial upgrading by improving factor productivity and total factor productivity [12]. The upgrading of industrial structure can inject new momentum into regional development, create new jobs, and promote high-quality local economic development. Since 2010, China's labor supply and demand pattern has shifted from the basic balance in total labor supply to an insufficient labor supply. The labor participation rate and the size of the working-age population have both declined [13]. The automation and efficiency improvement brought by technological progress are considered key to alleviating labor shortages and maintaining the potential growth rate of the economy, thereby improving the quality of economic development. Therefore, promoting technological progress to optimize factor distribution structure and promote regional development level has become an important issue in advancing Chinese-style modernization and ultimately achieving common prosperity [14]. This leads to the hypothesis:

H2: Technological progress has a positive effect on the level of regional development.

2.3 Aging and Economic Development Level

Population aging is an important problem and challenge facing China in the new stage of economic development [15, 16]. From 2000 to 2024, the old-age dependency ratio increased by nearly 10%. Based on the standard life-cycle model, the increase in the old-age dependency ratio implies a decline in the proportion of the working-age population, resulting in a shrinking labor supply, lower social production efficiency [17], and a decline in the social savings rate, thus inhibiting economic growth. At the same time, China's labor market has been widely discussed in relation to the Lewis turning point, as the decline in surplus labor has increased pressure on labor supply [18, 19]. From the perspective of induced technological change theory, labor scarcity may encourage firms to adopt machinery,

automation, and other labor-saving technologies to improve labor productivity. Therefore, technological progress may partially offset the negative effects of population aging on regional economic development, although this mechanism should be interpreted cautiously and supported by further empirical evidence [20, 21].

With the gradual increase in the proportion of urban elderly residents, demand for elderly-related resources, infrastructure, health care, and social services has continued to grow, giving rise to the development of the silver economy. This includes sectors such as pension finance, elderly cultural tourism, smart health care, daily care, housekeeping services, and age-friendly transformation, which may become new sources of local economic development [22]. At the same time, healthy older adults, especially those in the early stage of old age, may still represent an important form of human capital. Through a flexible retirement system [23] and elderly education [24], their experience, skills, and social participation can be further utilized, thereby helping to respond to population aging and potentially contributing to a second demographic dividend. This leads to the following hypothesis:

H3: Population aging has a positive effect on the level of regional development.

3 Model Construction

3.1 Variable Selection

3.1.1 Explained variables

In this paper, the level of regional development is measured by gross domestic product (GDP), and this variable is denoted as Y [25]. GDP is a comprehensive economic indicator that reflects the total scale of economic activities within a region over a given period. It is commonly used to measure regional economic development because it represents the economic output generated by all resident production units in a region. In general, a higher regional GDP indicates a higher level of economic development.

3.1.2 Explanatory variables

Education support (X_1) is proxied by fiscal expenditure on education [26]. Education support is one of the important factors in promoting regional development. Fiscal education expenditure refers to government financial investment in the education sector, including expenditure on schools, educational institutions, teachers' salaries, and educational facilities. The increase of education expenditure can improve the quality of education and increase human capital accumulation, thus promoting the long-term development of the regional economy. The level of education expenditure directly affects regional education development and talent cultivation, which in turn influences regional innovation capacity and competitiveness.

Technological progress (X_2) is measured by technology market turnover [27]. Technological progress is a key driving force of economic growth. Technology market turnover refers to the total transaction value between technology buyers and sellers in the technology market. This indicator can reflect the activity of regional technology transactions and the maturity of the technology market. It measures the circulation and value realization of technology as a marketable commodity. It also serves as an important bridge between technological innovation and real economic productivity, and can reflect the practical contribution of technology to economic activities. A higher technology market turnover generally indicates more active technological innovation and application in a region, as well as a stronger capacity to transform scientific and technological achievements into actual productivity.

However, this indicator may have a certain lag and may not fully capture independent R&D achievements or technological innovations within enterprises that have not yet entered the market [28]. At the same time, the turnover of the technology market may be affected by market cycles and policy subsidies, resulting in volatility. Nevertheless, it remains an effective and widely used proxy variable in measuring technology application and transformation efficiency.

Aging (X_3) is represented by the old-age dependency ratio [29], which refers to the ratio of the elderly population to the working-age population. An increase in this ratio may reduce labor supply in the labor market, while also increasing social demand for elderly care, social security, and health-care services. Therefore, population aging may have a dual impact on regional economic development.

3.2 Model Construction

The initial setting of the initial linear model is as follows.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon \quad (1)$$

where, Y is the explained variable, X_1 , X_2 , and X_3 are explanatory variables, β_0 is the intercept term, β_1 , β_2 , and β_3 are the coefficients of their respective variables, ε is the error term.

The ordinary least squares (OLS) method was used to estimate the model parameters in EVIEWS 13.0, and the economic implications were interpreted based on the estimated results.

3.3 Data Source

The data used in this study are macro-level data for Shandong Province from 2004 to 2022. All data were obtained from the China Statistical Yearbook [30] and the Shandong Statistical Yearbook [31].

4 Regression Analysis

4.1 Scatter Plot Analysis of Variable Relationships

To intuitively examine the relationships between the explanatory variables and the explained variable and to determine the appropriate functional form of the model, this study first draws scatter plots of Y against X_1 , X_2 , and X_3 , as shown in Figure 1. A strong positive linear association is observed between Y and X_1 , with the scatter points closely distributed around the fitted line. In contrast, the scatter plots of Y against X_2 and X_3 show certain nonlinear characteristics. This suggests that a simple linear model may not fully capture the complex relationships among the variables, and that logarithmic transformation or nonlinear model correction may be necessary in the subsequent analysis.

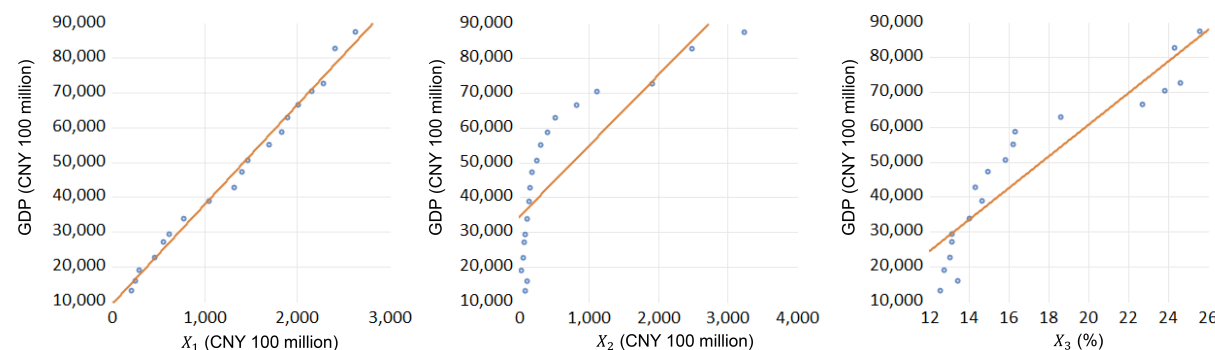


Figure 1. Scatter plots of Y against X_1 , X_2 , and X_3

Note: Y = gross domestic product (GDP); X_1 = education support; X_2 = technological progress; X_3 = population aging.

4.2 Preliminary Regression Results and Model Diagnosis

Based on the sample data from 2004 to 2022, this study first uses the OLS to estimate the specified linear model, and the results are presented in Table 1. The model has a high overall goodness of fit, with $R^2 = 0.994$ and adjusted $R^2 = 0.993$, indicating that the explanatory variables have strong explanatory power for the variation in the explained variable. The p -value of the F -test is less than 0.001, indicating that the regression equation is statistically significant as a whole. However, X_2 and X_3 do not pass the significance test at the 5% level. In addition, the relatively large constant term may indicate that the model specification requires further adjustment. Therefore, the preliminary results suggest that the original linear model may have specification problems and should be further corrected.

Table 1. Ordinary least squares (OLS) regression results for the initial linear model

Variable	Coefficient	SE	t -Statistic	p -Value
Constant	9,975.761	3,799.461	2.626	0.0191
Education support (X_1)	26.186	1.373	19.067	<0.001
Technological progress (X_2)	2.313	1.105	2.093	0.054
Population aging (X_3)	64.716	324.299	0.200	0.845
Statistic	Value	Statistic	Value	
R^2	0.994	Mean dependent variable	47,323.970	
Adjusted R^2	0.993	SD of dependent variable	22,901.790	
SE of regression	1,911.181	Akaike information criterion	18.133	
Residual sum of squares	54,789,198	Schwarz criterion	18.332	
Log likelihood	-168.268	Hannan-Quinn criterion	18.167	
F -statistic	856.561	Durbin-Watson statistic	1.555	
p -value (F -statistic)	<0.001			

Note: SE = standard error; SD = standard deviation; R^2 = coefficient of determination.

4.3 Model Correction and Double Logarithmic Regression Analysis

Based on the preliminary regression results and the scatter plot analysis, this study modifies the original specification and adopts a double-logarithmic regression model. Specifically, the natural logarithms of all variables are taken, and the following model is constructed:

$$\ln Y = \beta_0 + \beta_1 \ln X_1 + \beta_2 \ln X_2 + \beta_3 \ln X_3 + \varepsilon \quad (2)$$

The sample data from 2004 to 2022 are estimated using the OLS method and the regression results are shown in Table 2.

Table 2. Ordinary least squares (OLS) regression results for the double-logarithmic model

Variable	Coefficient	SE	<i>t</i> -Statistic	<i>p</i> -Value
Constant	5.328	0.308	17.273	<0.001
ln <i>X</i> ₁	0.615	0.024	25.974	<0.001
ln <i>X</i> ₂	−0.020	0.031	−0.654	0.523
ln <i>X</i> ₃	0.408	0.156	2.620	0.019
Statistic	Value	Statistic	Value	
<i>R</i> ²	0.995	Mean dependent variable	10.629	
Adjusted <i>R</i> ²	0.994	SD of dependent variable	0.569	
SE of regression	0.043	Akaike information criterion	−3.261	
Residual sum of squares	0.028	Schwarz criterion	−3.062	
Log likelihood	34.981	Hannan-Quinn criterion	−3.228	
<i>F</i> -statistic	1,034.365	Durbin-Watson statistic	1.439	
<i>p</i> -value (<i>F</i> -statistic)	<0.001			

Note: SE = standard error; SD = standard deviation; R^2 = coefficient of determination; X_1 = education support; X_2 = technological progress; X_3 = population aging.

The corrected model has a strong fitting effect. The adjusted coefficient of determination is adjusted $R^2 = 0.994$, indicating that the model fits the sample data very well. The *p*-value corresponding to the *F*-statistic is less than 0.001, indicating that the regression equation is highly significant as a whole. At the 5% significance level, the critical value is approximately $t_{0.025}(15) = 2.13$. The *t*-statistic of $\ln X_1$ is 25.974, with $p < 0.001$, indicating that education expenditure passes the significance test. Its coefficient is 0.615, which means that a 1% increase in education expenditure is associated with an average increase of approximately 0.615% in regional GDP. The *t*-statistic of $\ln X_3$ is 2.620, with $p = 0.019$, indicating that the old-age dependency ratio also passes the significance test at the 5% level. In contrast, the *t*-statistic of $\ln X_2$ is −0.654, with $p = 0.523$, indicating that technology market turnover does not pass the significance test in the current model. The insignificance of $\ln X_2$ may be related to the lagged effect of technology market turnover on economic growth or to potential multicollinearity with other variables. Although the overall goodness of fit of the model is very high, the insignificant coefficient of $\ln X_2$ suggests that the role of technological progress should be interpreted cautiously in the current specification. In addition, the Durbin-Watson statistic is 1.439, suggesting that possible autocorrelation should be further examined. Therefore, subsequent analysis may consider eliminating insignificant variables, introducing lag terms, or conducting additional diagnostic tests to further optimize the model.

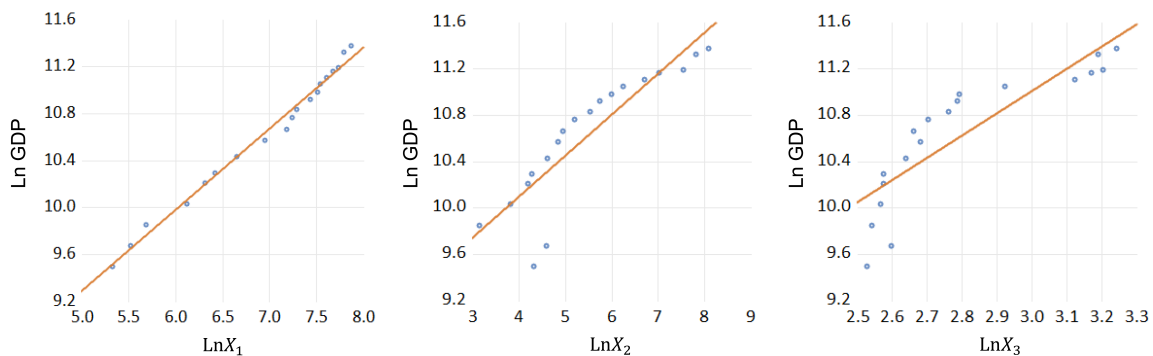


Figure 2. Scatter plots of $\ln X_1$, $\ln X_2$, $\ln X_3$ and $\ln Y$

Note: Y = gross domestic product (GDP); X_1 = education support; X_2 = technological progress; X_3 = population aging.

The relationship between $\ln Y$ and each logarithmic explanatory variable is further examined in Figure 2. The scatter points of $\ln X_1$ and $\ln X_3$ are closely distributed around the fitted lines, indicating strong linear relationships with $\ln Y$ after logarithmic transformation. This is consistent with the significant regression results for $\ln X_1$ and $\ln X_3$. By contrast, the scatter points of $\ln X_2$ are relatively more dispersed, especially in the lower-value range, which may help explain why $\ln X_2$ is statistically insignificant in the regression model.

4.4 Heteroscedasticity Test

Although the double-logarithmic model constructed above shows good overall fit and statistical significance, heteroscedasticity may still exist in the error terms. If heteroscedasticity is present, the OLS estimators may remain unbiased under the relevant assumptions, but they are no longer efficient, and the standard errors, t -statistics, and F -statistics may become unreliable. Therefore, this study uses the White test to examine whether heteroscedasticity exists in the double-logarithmic model. The test results are shown in Table 3.

Table 3. White test results for heteroscedasticity

Test Statistic	Value	p -Value
F -statistic	1.752	0.208
$\text{Obs} \cdot R^2$	12.096	0.208
Scaled explained SS	2.696	0.975

Note: $\text{Obs} \cdot R^2$ = number of observations multiplied by coefficient of determination; SS = sum of squares.

Based on the chi-square statistic, $\text{Obs} \cdot R^2 = 12.096$, which is smaller than the critical value $\chi_{0.05}^2(9) = 16.919$. In addition, the corresponding p -value is 0.208, which is greater than 0.05. Therefore, the null hypothesis of homoscedasticity cannot be rejected. The White test results indicate that the double-logarithmic model does not show significant heteroscedasticity at the 5% significance level. Thus, no additional correction such as weighted least squares (WLS) is required for heteroscedasticity.

4.5 Sequence Autocorrelation Test

In time-series regression analysis, serial autocorrelation in the error terms may reduce the efficiency of the model estimates and affect statistical inference. The Durbin-Watson statistic of the double-logarithmic model is 1.439. According to the Durbin-Watson critical values at the 5% significance level, with $k = 3$ explanatory variables and $n = 19$ observations, the lower bound is $d_L = 1.08$ and the upper bound is $d_U = 1.54$. Since the Durbin-Watson statistic lies between d_L and d_U , the Durbin-Watson test result is inconclusive rather than providing clear evidence of positive autocorrelation.

To obtain a more reliable conclusion, this study further applies the Breusch-Godfrey Lagrange multiplier (LM) test to examine serial autocorrelation. The results are shown in Table 4.

Table 4. Breusch-Godfrey Lagrange multiplier (LM) test results for serial autocorrelation

Test Statistic	Value	p -Value
F -statistic	1.552	0.249
$\text{Obs} \cdot R^2$	3.663	0.160

Note: $\text{Obs} \cdot R^2$ = number of observations multiplied by coefficient of determination.

Table 4 reports the LM test results with two lags. The null hypothesis is that there is no serial autocorrelation up to the specified lag order. The $\text{Obs} \cdot R^2$ statistic is 3.663, which is smaller than the critical value $\chi_{0.05}^2(2) = 5.991$. The corresponding p -value is 0.160, which is greater than 0.05. Therefore, the null hypothesis cannot be rejected. The results indicate that the double-logarithmic model does not have significant serial autocorrelation at the 5% significance level.

4.6 Multicollinearity Test

Although the double-logarithmic model has a high goodness of fit and the overall F -test is significant, $\ln X_2$ does not pass the t -test. This may indicate potential multicollinearity among the explanatory variables. Therefore, this study first examines the correlation coefficient matrix of the explanatory variables. The results are shown in Table 5.

The correlation coefficients among the explanatory variables are relatively high. In particular, the correlation coefficient between $\ln X_2$ and $\ln X_3$ reaches 0.967, indicating a strong linear relationship between these two variables. This preliminarily suggests the existence of multicollinearity.

Table 5. Correlation coefficient matrix of explanatory variables

Variable	$\ln X_1$	$\ln X_2$	$\ln X_3$
$\ln X_1$	1.000		
$\ln X_2$	0.851 (<0.001)	1.000	
$\ln X_3$	0.820 (<0.001)	0.967 (<0.001)	1.000

Note: X_1 = education support; X_2 = technological progress; X_3 = population aging. Values in parentheses are p -values.

To further evaluate the severity of multicollinearity, the variance inflation factor (VIF) method is used. It is generally considered that a centered VIF value greater than 10 indicates serious multicollinearity. The results are shown in Table 6. The centered VIF values of $\ln X_2$ and $\ln X_3$ are greater than 10, indicating serious multicollinearity between these variables. Combined with the correlation coefficient matrix, the results suggest that multicollinearity may be one reason why $\ln X_2$ is statistically insignificant in the double-logarithmic model. Therefore, further model correction is necessary.

Table 6. Variance inflation factor (VIF) test results

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
Constant	0.095	968.412	Not available
$\ln X_1$	0.001	278.519	3.628
$\ln X_2$	0.001	306.269	18.179
$\ln X_3$	0.024	1,953.460	15.329

Note: X_1 = education support; X_2 = technological progress; X_3 = population aging.

4.7 Correction of Multicollinearity

In view of the multicollinearity diagnosed above, this study uses a stepwise regression approach to modify the model. Since $\ln X_2$ is highly correlated with $\ln X_3$ and is not statistically significant in the previous regression, $\ln X_2$ is removed from the model, while $\ln X_1$ and $\ln X_3$ are retained. The corrected model is specified as follows:

$$\ln Y = \beta_0 + \beta_1 \ln X_1 + \beta_3 \ln X_3 + \varepsilon \quad (3)$$

The regression results of the corrected model are shown in Table 7.

Table 7. Regression results after correcting multicollinearity

Variable	Coefficient	SE	t -Statistic	p -Value
Constant	5.515	0.110	50.040	<0.001
$\ln X_1$	0.609	0.021	28.523	<0.001
$\ln X_3$	0.317	0.068	4.640	<0.001
Statistic	Value	Statistic	Value	
R^2	0.995	Adjusted R^2	0.994	
F -statistic	1,608.928	p -value (F -statistic)	<0.001	
Durbin-Watson statistic	1.497			

Note: SE = standard error; R^2 = coefficient of determination; X_1 = education support; X_3 = population aging.

After removing $\ln X_2$, the t -statistics of $\ln X_1$ and $\ln X_3$ are 28.523 and 4.640, respectively, and both variables are statistically significant at the 1% level. The F -statistic is 1,608.928, with $p < 0.001$, indicating that the corrected model is statistically significant as a whole. The adjusted R^2 is 0.994, suggesting that the model still maintains a very high goodness of fit.

Table 8 reports the VIF test results for the corrected model. The centered VIF values of $\ln X_1$ and $\ln X_3$ are both below 10, indicating that the multicollinearity problem has been effectively reduced.

In addition, the corrected model is further examined using the White test and the Breusch-Godfrey LM test. The test results show that the corresponding p -values are greater than 0.05, indicating that the corrected model does not show significant heteroscedasticity or serial autocorrelation. Therefore, the final estimated regression equation is:

$$\ln Y = 5.515 + 0.609 \ln X_1 + 0.317 \ln X_3 \quad (4)$$

Table 8. Revised variance inflation factor (VIF) test results

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
Constant	0.012	128.259	Not available
$\ln X_1$	0.000	234.787	3.058
$\ln X_3$	0.005	389.711	3.058

Note: X_1 = education support; X_3 = population aging.

4.8 Robustness Test

To verify the reliability of the benchmark regression results, this study conducts a robustness test by replacing the explained variable with per capita GDP and re-estimating the double-logarithmic model. The results are shown in Table 9. The coefficients of the core explanatory variables $\ln X_1$ and $\ln X_3$ are both significantly positive at the 1% level, and their signs are consistent with those in the baseline regression. This indicates that the main conclusion does not depend entirely on the specific measurement of the explained variable and has a certain degree of robustness.

Furthermore, diagnostic tests also support the statistical reliability of this robustness model. The F -statistic is 1,417.623 ($p < 0.001$), indicating that the overall regression equation is highly significant. The Durbin-Watson statistic is 1.384; although this is slightly below the ideal value of 2, it remains within an acceptable range for small-sample time-series data with a clear trend, suggesting that the model does not show strong evidence of first-order autocorrelation. Additionally, the high adjusted R^2 (0.994) demonstrates the model maintains strong explanatory power even after replacing the dependent variable.

Table 9. Robustness test results using per capita GDP as an alternative dependent variable

Variable	Coefficient	SE	<i>t</i> -Statistic	<i>p</i> -Value
Constant	−3.267	0.109	−29.886	<0.001
ln <i>X</i> ₁	0.579	0.021	27.339	<0.001
ln <i>X</i> ₃	0.251	0.068	3.700	<0.005
Statistic	Value	Statistic	Value	
<i>R</i> ²	0.994	Adjusted <i>R</i> ²	0.994	
<i>F</i> -statistic	1,417.623	<i>p</i> -value (<i>F</i> -statistic)	<0.001	
Durbin-Watson statistic	1.384			

Note: GDP = gross domestic product; SE = standard error; R^2 = coefficient of determination; X_1 = education support; X_3 = population aging.

5 Conclusions and Recommendations

5.1 Conclusions

Based on time-series data for Shandong Province from 2004 to 2022, this study empirically examines the relationships among education support, technological progress, population aging, and regional economic development. After diagnosing multicollinearity in the initial model, technology market turnover was removed through stepwise regression, and a corrected model including education support and population aging was constructed. The main conclusions are as follows.

Education support is positively associated with regional economic development. The empirical results show that fiscal education expenditure has a significant positive relationship with regional GDP, and its estimated elasticity is relatively large. This suggests that increasing education investment may contribute to human capital accumulation, improve labor productivity, and support long-term regional economic development.

Population aging also shows a significant positive association with regional economic development in the corrected model. This finding differs from the traditional view that aging necessarily constrains economic growth. One possible explanation is that Shandong Province has a relatively high level of urbanization and household income, which may provide favorable conditions for the development of the silver economy. Growing demand for elderly care, health-care services, elderly consumer goods, and related industries may create new sources of local economic activity. In addition, some healthy older adults may continue to contribute to economic development through delayed retirement, re-employment, entrepreneurship, or social participation.

Additionally, the technological progress variable, measured by technology market turnover, did not pass the significance test in the initial double-logarithmic model and was later removed due to serious multicollinearity. Therefore, the direct effect of technological progress on regional economic development is not fully verified in this study and should be further examined in future research.

5.2 Recommendations

5.2.1 Prioritize education development and optimize the allocation of urban and regional education resources

The government should continue to increase financial investment in education while paying greater attention to spatial equity and structural optimization. At the urban level, high-quality educational resources should be extended to new urban areas, suburban areas, and densely populated communities to reduce unequal access to education. At the regional governance level, provincial coordination should be strengthened, and more educational resources should be allocated to rural, remote, underdeveloped, and minority areas through transfer payments and targeted support. These measures can help narrow regional education gaps, promote the equalization of basic public services, and provide long-term human capital support for economic development. In addition, vocational education should be better connected with leading regional industries in order to cultivate high-quality technical and skilled workers for industrial transformation and upgrading.

5.2.2 Develop the silver economy from the perspective of urban governance

From the perspective of urban governance, population aging should be actively addressed and transformed into a potential source of development. First, urban elderly-care infrastructure should be improved. Age-friendly design should be systematically incorporated into urban planning, including community-based elderly-care service centers, barrier-free public facilities, and smart elderly-care service platforms. These measures can help transform elderly-care demand into real economic activity.

Second, regional silver economy clusters should be cultivated. Relying on the medical, scientific, educational, and service resources of central cities such as Jinan and Qingdao, Shandong Province can develop industrial clusters integrating medical rehabilitation, elderly-care services, elderly cultural tourism, finance, and insurance. These clusters may also promote coordinated development in surrounding cities.

Third, a flexible retirement system and re-employment support mechanisms for older adults should be explored. An information database and re-employment service platform for healthy older adults could be established to encourage their participation in voluntary services, technical consultation, community governance, and other social and economic activities. This may help relieve labor supply pressure, improve the utilization of human capital, and promote the development of a potential silver dividend.

5.3 Research Limitations and Future Prospects

Although this study has drawn some meaningful findings, there are some limitations. Firstly, the sample size of the study is relatively limited, covering only 19 years of provincial time-series data. A small sample may affect the stability of model estimation and limit the generalizability of the conclusions. Second, the study uses provincial aggregate data, which may not fully capture differences among cities within Shandong Province. Third, some variables, such as technological progress, may have lagged effects that are not fully reflected in the current model.

Future research can use panel data from prefecture-level cities in Shandong Province to expand the sample size and improve the reliability of the results. Variables such as GDP, fiscal education expenditure, and technology market turnover may contain common time trends or inflation effects. Future research should conduct unit root tests, cointegration tests, and constant-price adjustments for monetary variables to reduce the risk of spurious regression and improve the robustness of the findings.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the author used AI solely for the purpose of language editing, grammar checking, and improving readability. The author confirms that all intellectual content, including the research design, data analysis, interpretation of results, and conclusions, was generated entirely by the author. The author has reviewed and verified all AI-assisted edits to ensure accuracy and originality, and the author accept full responsibility for the final content of this work.

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